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Score: _____

Directions: Do problem 2, and one of problems 1 or 3.

1. Let H be a subgroup of a group G with the property that $ghg^{-1} \in H$ for every $g \in G$ and $h \in H$. Prove that $gH = Hg$ for every $g \in G$. (i.e., show that the left and right cosets of H are identical.)

Proof (Direct) Suppose H has the property that $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$. To show $gH = Hg$, we will show (A) $gH \subseteq Hg$, and (B) $Hg \subseteq gH$ (for any g). Let $g \in G$.

(A) Suppose $x \in gH$. This means $x = gh$ for some $h \in H$. Then $xg^{-1} = ghg^{-1} \in H$. Hence $x = (ghg^{-1})g$ where $ghg^{-1} \in H$, and this means $x \in Hg$. Therefore $gH \subseteq Hg$.

(B) Suppose $x \in Hg$. This means $x = hg$ for some $h \in H$. Then $g^{-1}x = g^{-1}hg = (g^{-1})h(g^{-1})^{-1} \in H$. From this $x = g((g^{-1})h(g^{-1})^{-1})$ with $(g^{-1})h(g^{-1})^{-1} \in H$, so $x \in gH$. Therefore $Hg \subseteq gH$.

Together (A) and (B) imply $gH = Hg$. $\square$

2. List the cosets of the subgroup $H = \langle (0,1) \rangle \subseteq \mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0,0), (1,0), (0,1), (1,1)\}$

$$H = \langle (0,1) \rangle = \{(0,0), (0,1)\}$$

$$(1,0) + H = \{(1,0), (1,1)\}$$

$$(0,1) + H = \{(0,1), (0,0)\}$$

$$(1,1) + H = \{(1,1), (1,0)\}$$

Only two
cosets

H and

$(1,0) + H$

3. Suppose $g^n = e$ for some element g of a group G , and positive integer n .

Prove that the order of g divides n . (Consider using the division algorithm.)

Proof (Direct) Suppose $g^n = e$.

Let the order of g be k . This means k is the smallest positive integer for which $g^k = e$.

By the division algorithm, $n = qk + r$ for some $q \geq 0$ and $0 \leq r < k$. Then

$$e = g^n = g^{qk+r} = (g^k)^q g^r = e^q g^r = g^r.$$

In other words $g^r = e$, and $0 \leq r < k$.

But k is the smallest positive integer for which $g^k = e$, and $g^r = e$ where $r < k$.

Therefore $r = 0$, and $n = qk + r = qk$,

so k divides n . $\blacksquare$